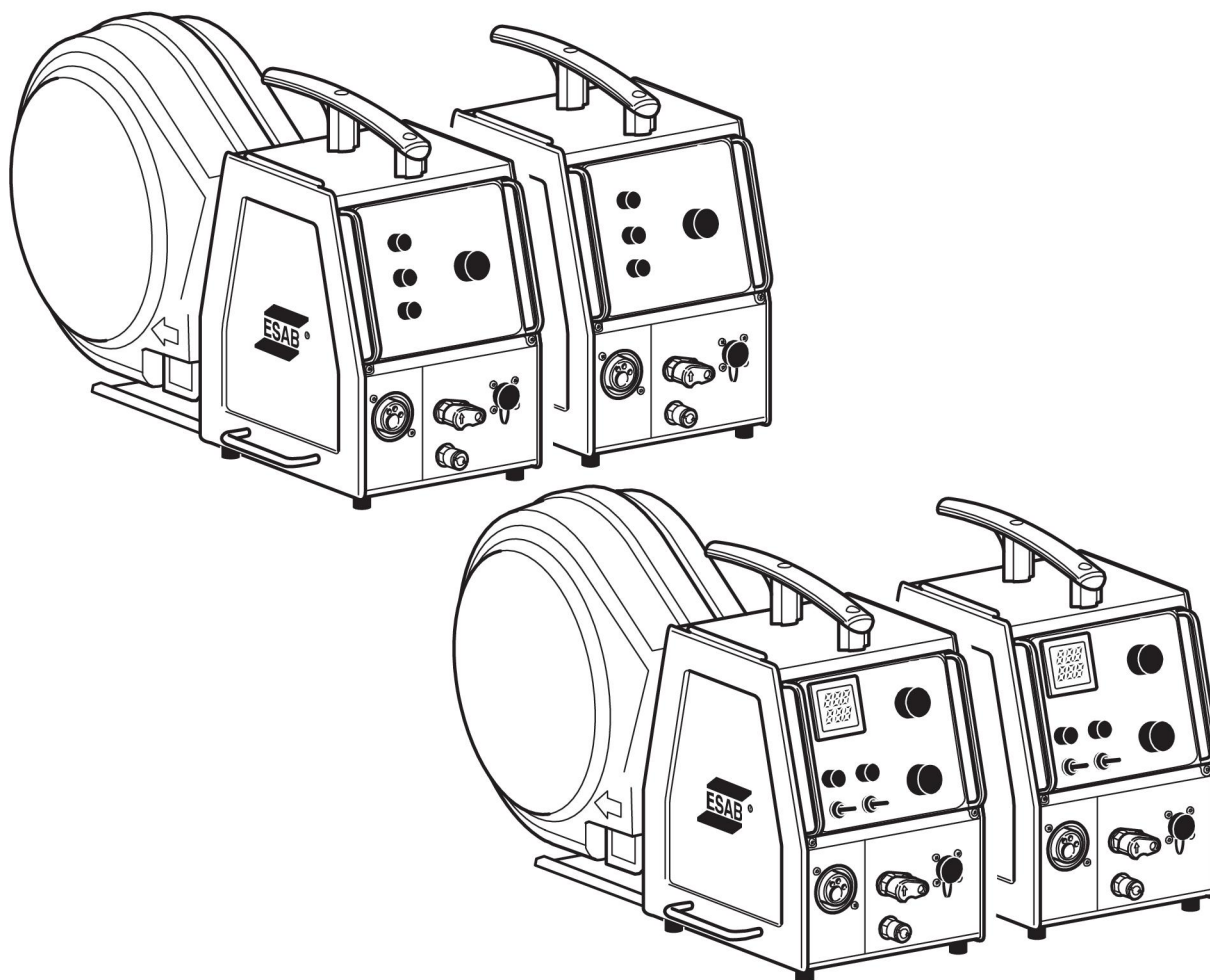


Origo™

Feed 304 M12, Feed 304 M13, Feed 484 M12, Feed 484 M13



Service manual

READ THIS FIRST	3
INTRODUCTION	4
TECHNICAL DATA	5
WIRING DIAGRAM	7
Component description	7
Feed 304 M12 up to and including serial no. 620-xxx-xxxx,	9
Feed 484 M12 all serial numbers	
Feed 304 M12 serial no. 917-xxx-xxxx	11
Feed 304 M12 from serial no. 623-xxx-xxxx	13
Feed 304 M13 up to and including serial no. 620-xxx-xxxx,	15
Feed 484 M13 all serial numbers	
Feed 304 M13 serial no. 917-xxx-xxxx	17
Feed 304 M13 from serial no. 623-xxx-xxxx	19
DESCRIPTION OF OPERATION	21
13AP1 Control board	21
13AP1:1 Power supply	21
13AP1:2 Activation, power source	22
13AP1:3 Crater filling	22
13AP1:4 Voltage adjustment	23
13AP1:5 Gas valve	24
13AP1:6 2-stroke / 4-stroke	24
13AP1:7 Burn-back time	25
13AP1:8 Wire feed speed	25
13AP1:9 Motor driving / braking	26
13AP1:10 Tachometer input	27
13AP1:11 Current relay, Creep start / Normal start	28
13AP1:12 Start / Stop	28
13AP1:13 Processor, Starting sequence, Motor type	29
13AP1:14 Gas purge	30
13AP1:15 Inching	30
13AP1:16 Remote control input	30
13AP1 Component positions	32
13AP2 Digital display	34
SERVICE INSTRUCTIONS	37
What is ESD?	37
Spare parts for connection sets, 23 pole	38
SPARE PARTS	39

READ THIS FIRST

Maintenance and repair work must be performed by an experienced person, and electrical work only by a trained electrician. Use only recommended replacement parts.

This service manual is intended for use by technicians with electrical/electronic training for help in connection with fault-tracing and repair.

Use the wiring diagram as a form of index for the description of operation. The circuit boards are divided into numbered blocks, which are described individually in more detail in the description of operation. Component names in the wiring diagram are listed in the component description.

Use the spare parts list as a guide to where the components are located in the equipment. The spare parts list is published as a separate document, see the "SPARE PARTS" chapter in this manual.

This manual contains details of design changes that have been made up to and including February 2017.

The manual is valid for:

ESABFeed 30-4 M12, ESABFeed 30-4 M13, ESABFeed 48-4 M12 and ESABFeed 48-4 M13 with serial no. 238-xxx-xxxx, 410-xxx-xxxx, 440-xxx-xxxx, 502-xxx-xxxx

Feed 304 M12 and Feed 304 M13 with serial no. 620-xxx-xxxx, 917-xxx-xxxx, 623-xxx-xxxx

Feed 484 M12 and Feed 484 M13 with serial no. 620-xxx-xxxx



CAUTION!

Read and understand the instruction manual before installing or operating.



The ESABFeed 30-4 M12, ESABFeed 30-4 M13, ESABFeed 48-4 M12, ESABFeed 48-4 M13, Feed 304 M12, Feed 304 M13, Feed 484 M12 and Feed 484 M13 are designed and tested in accordance with international and European standards IEC/EN 60974. On completion of service or repair work, it is the responsibility of the person(s) performing the work to ensure that the product still complies with the requirements of the above standard.

INTRODUCTION

The wire feeders are renamed:

New name

Origo™ Feed 304, M12

Origo™ Feed 484, M12

Origo™ Feed 304, M13

Origo™ Feed 484, M13

Old name

ESABFeed 30-4, M12

ESABFeed 48-4, M12

ESABFeed 30-4, M13

ESABFeed 48-4, M13



*The wire feed units with control panel **M12** are intended for MIG/MAG-welding together with stepped welding power sources.*



*The wire feed units with control panel **M13** are intended for MIG/MAG-welding together with infinitely adjustable welding power sources.*

TECHNICAL DATA

	Feed 304	Feed 484
Mains voltage	42 V, 50–60 Hz	42 V, 50–60 Hz
Power requirement	336 VA	378 VA
Motor current $I_{\max}$	8 A	9 A
Control panel M12, settings data		
Wire feed speed	1.9–25.0 m/min (6.2–82.0 ft/min)	1.9–25.0 m/min (6.2–82.0 ft/min)
Burnback time	0–0.5 s	0–0.5 s
Creep start	OFF or ON	OFF or ON
2-/4-stroke	2-stroke or 4-stroke	2-stroke or 4-stroke
Control panel M13, settings data		
Wire feed speed	1.9–25.0 m/min (6.2–82.0 ft/min)	1.9–25.0 m/min (6.2–82.0 ft/min)
Burnback time	0–0.5 s	0–0.5 s
Crater filling time	0–5.1 s	0–5.1 s
Creep start	OFF or ON	OFF or ON
2-/4-stroke	2-stroke or 4-stroke	2-stroke or 4-stroke
Torch connection	EURO	EURO
Max. diameter wire bobbin	300 mm / *440 mm (11.8 in. / *17.3 in.)	300 mm / *440 mm (11.8 in. / *17.3 in.)
Wire dimension	0.6–1.6 mm (0.02–0.06 in.)	0.6–2.4 mm (0.02–0.09 in.)
Weight		
Basic version	11.5 kg (25.3 lb)	14.5 kg (32.0 lb)
With sealed bobbin holder	15.0 kg (33.0 lb)	19.0 kg (41.9 lb)
Dimensions (l × w × h)		
Basic version	380 × 275 × 400 mm (15.0 × 10.8 × 15.7 in.)	380 × 275 × 400 mm (15.0 × 10.8 × 15.7 in.)
With sealed bobbin holder	690 × 275 × 420 mm (27.2 × 10.8 × 16.5 in.)	690 × 275 × 420 mm (27.2 × 10.8 × 16.5 in.)
Operating temperature	-10 to +40 °C (14 to 104 °F)	-10 to +40 °C (14 to 104 °F)
Shielding gas	All types intended for MIG/MAG welding	All types intended for MIG/MAG welding
max pressure	0.5 MPa (5 bar)	0.5 MPa (5 bar)
Coolant	ESAB ready mixed coolant	ESAB ready mixed coolant
max pressure	0.5 MPa (5 bar)	0.5 MPa (5 bar)
Maximum permissible load at		
60% duty cycle	630 A	630 A
Enclosure class	IP23	IP23
With wire bobbin *440 mm	IP2X	IP2X

*Accessories can be found in the Instruction manual in the chapter "ACCESSORIES".

Duty cycle

The duty cycle refers to the time as a percentage of a ten-minute period that you can weld or cut at a certain load without overloading. The duty cycle is valid for 40 °C / 104 °F, or below.

Enclosure class

The **IP** code indicates the enclosure class, i.e. the degree of protection against penetration by solid objects or water.

Equipment marked **IP23** is intended for indoor and outdoor use.

Equipment marked **IP2X** is intended for indoor use.

WIRING DIAGRAM

Component description

13AP1	Main circuit board, with the control electronics (see the description in section <i>"13AP1 Control board"</i> , page 21)
13AP2	Circuit board with display, see the description in <i>"13AP2 Digital display"</i> , page 34. Only for ESABFeed with M13 panel.
13C2 - 13C6	Capacitor, 0.1 μ F.
13CXX	Capacitor, 4700 μ F. Must be fitted when the wire feed unit is used under difficult applications (e.g. low mains voltage, long cables etc.).
13G1	Feed 484: All serial numbers. Tachogenerator incorporated in motor 13M2. Feed 304: Up to and including serial no. 620-xxx-xxxx. Tachogenerator incorporated in motor 13M1. From serial no. 917-xxx-xxxx Pulsegenerator (encoder) incorporated in motor 13M1. See <i>"13AP1:10 Tachometer input"</i> , page 27.
13M1	Motor, rated voltage 42 V. Feed 304.
13M2, 13M3	Motor, rated voltage 12 V. Feed 484.
13RP1	Potentiometer, 10 kOhm, for setting the wire feed speed.
13RP2	Potentiometer, 10 kOhm, for setting the welding voltage. Only ESABFeed with M13 panel.
13RP3	Potentiometer, 10 kOhm, for setting the burn-back time.
13RP4	Potentiometer, 10 kOhm, with incorporated switch, for setting the crater fill time. Only ESABFeed with M13 panel.
13S1	Microswitch, monitoring the water connection. The pump in the water cooling unit starts when the switch is closed.
13SA1	Switch, 2/4-stroke changeover. When the switch is closed, 4-stroke control mode is selected.
13SA2	Switch, creep start function ON/OFF. When the switch is open, creep start function is operative.
13XP1	Panel plug, for connection to the welding power source.
13XP2	Panel plug, for welding current connection from the power source.
13XS...	Connectors with the designation 13XS... are socket connectors.
13XS13	23 pole panel socket for remote control.
13YV1	Gas valve



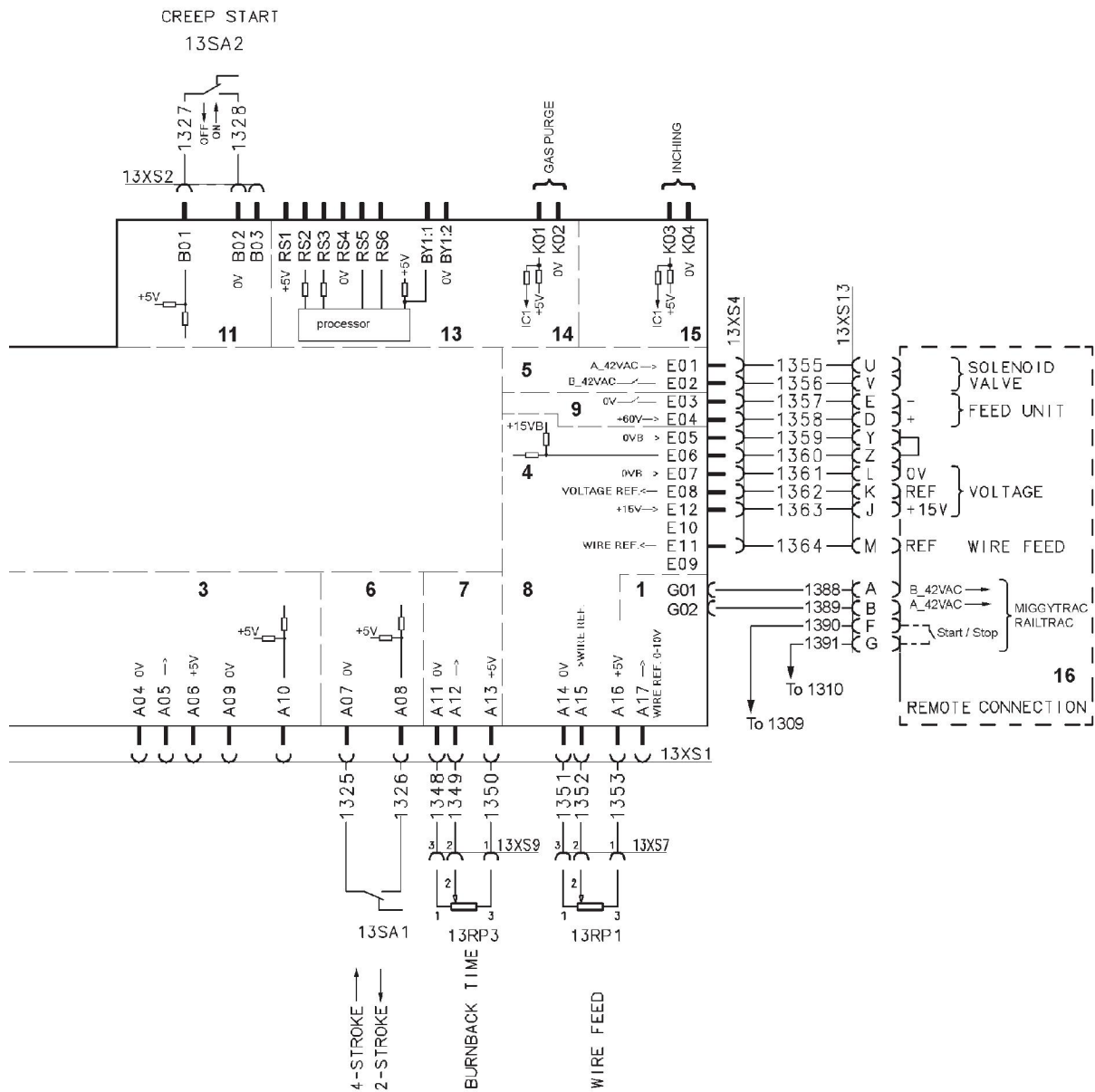
CAUTION!

STATIC ELECTRICITY can damage circuit boards and electronic components.

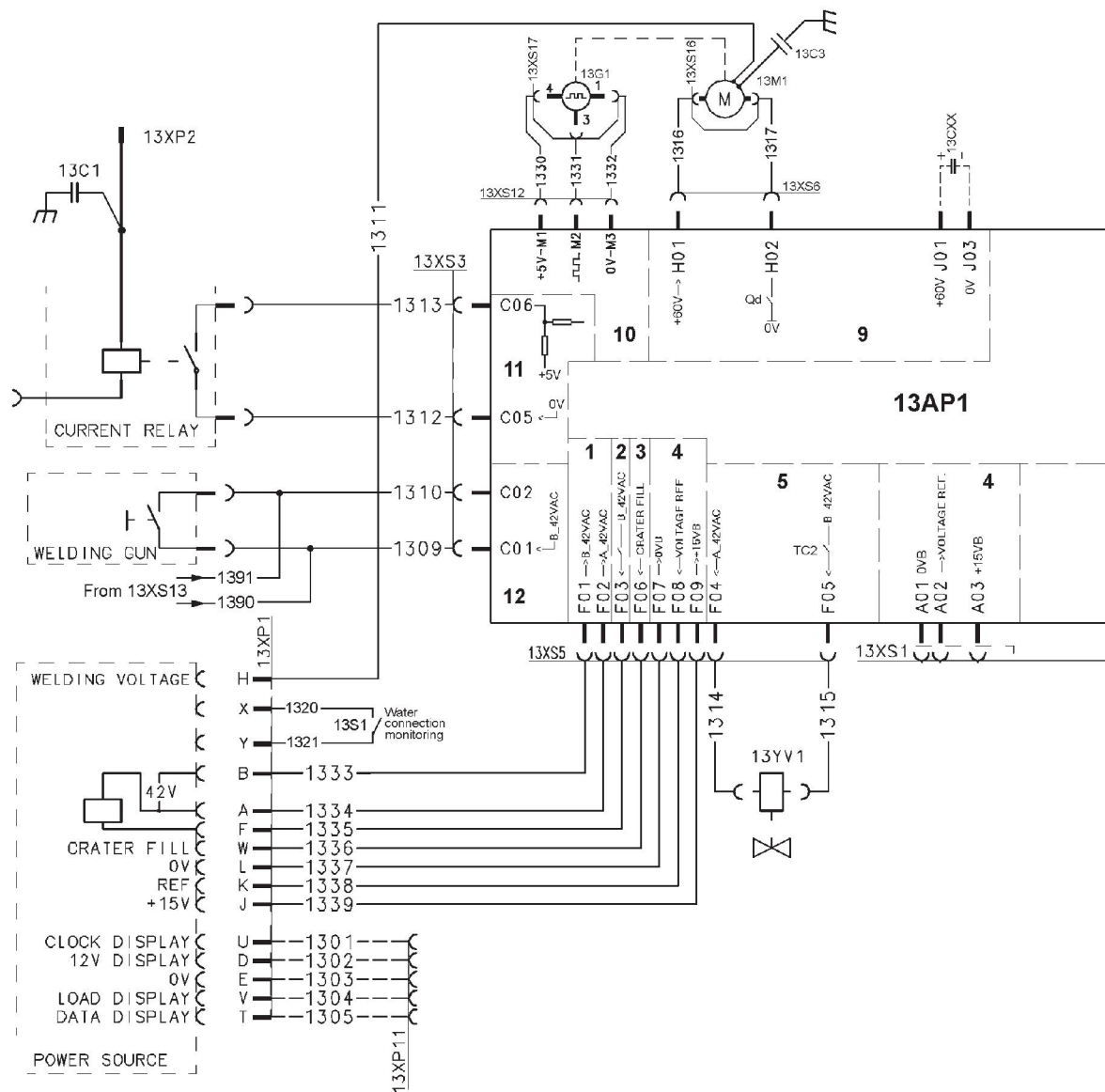
- Observe precautions for handling electrostatic sensitive devices.
- Use proper static-proof bags and boxes.

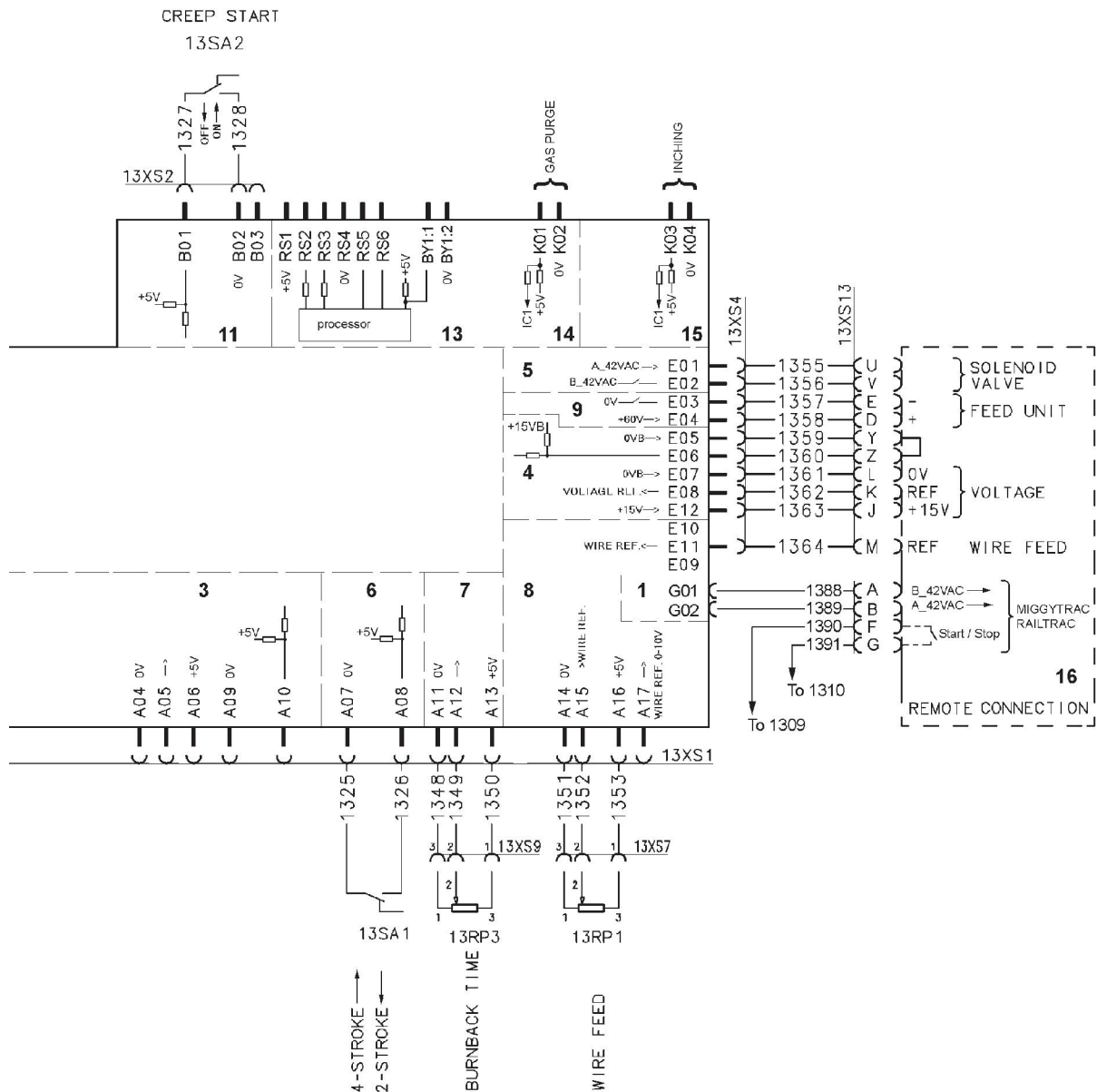


ESD

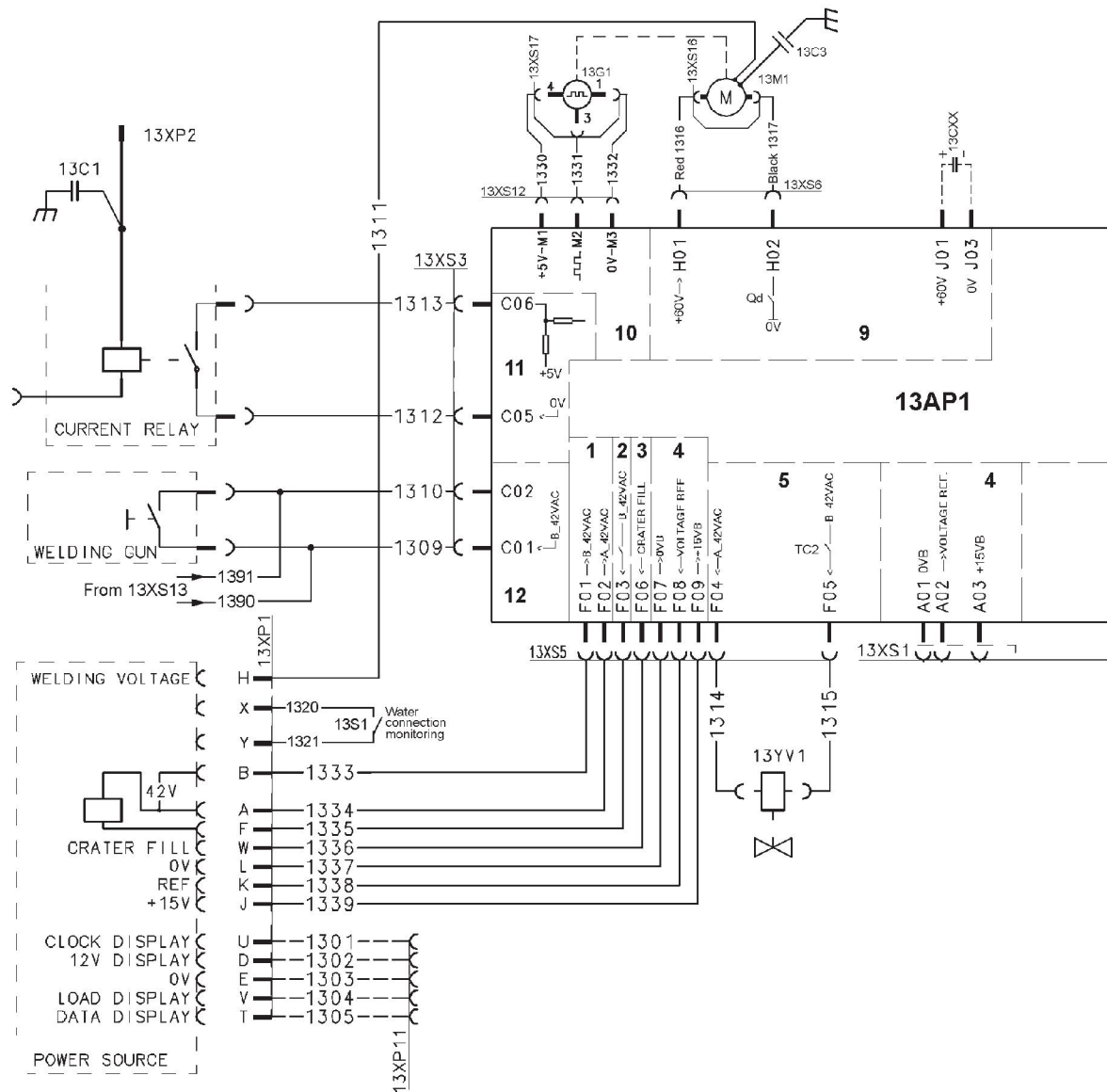


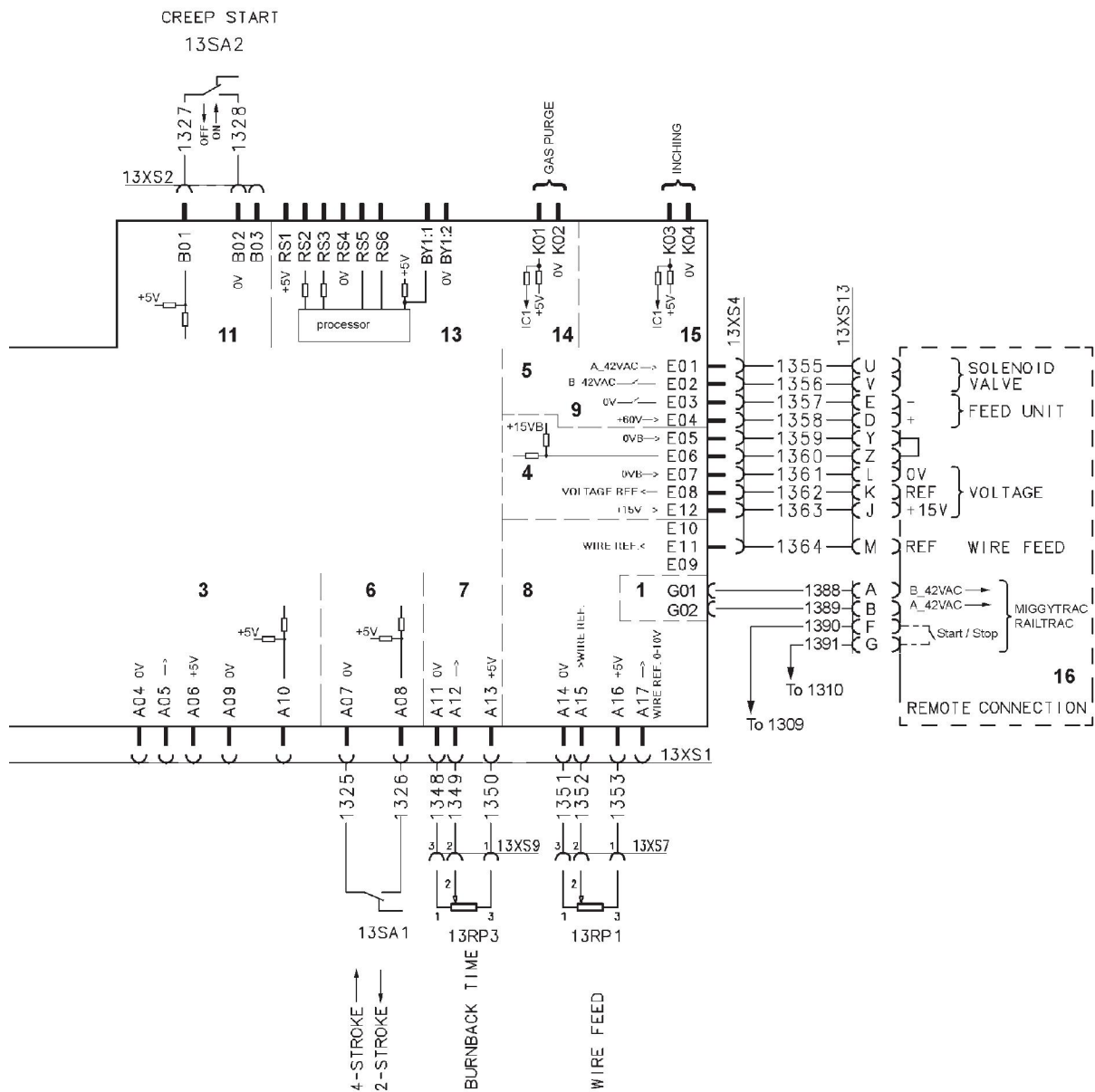
Feed 304 M12 serial no. 917-xxx-xxxx



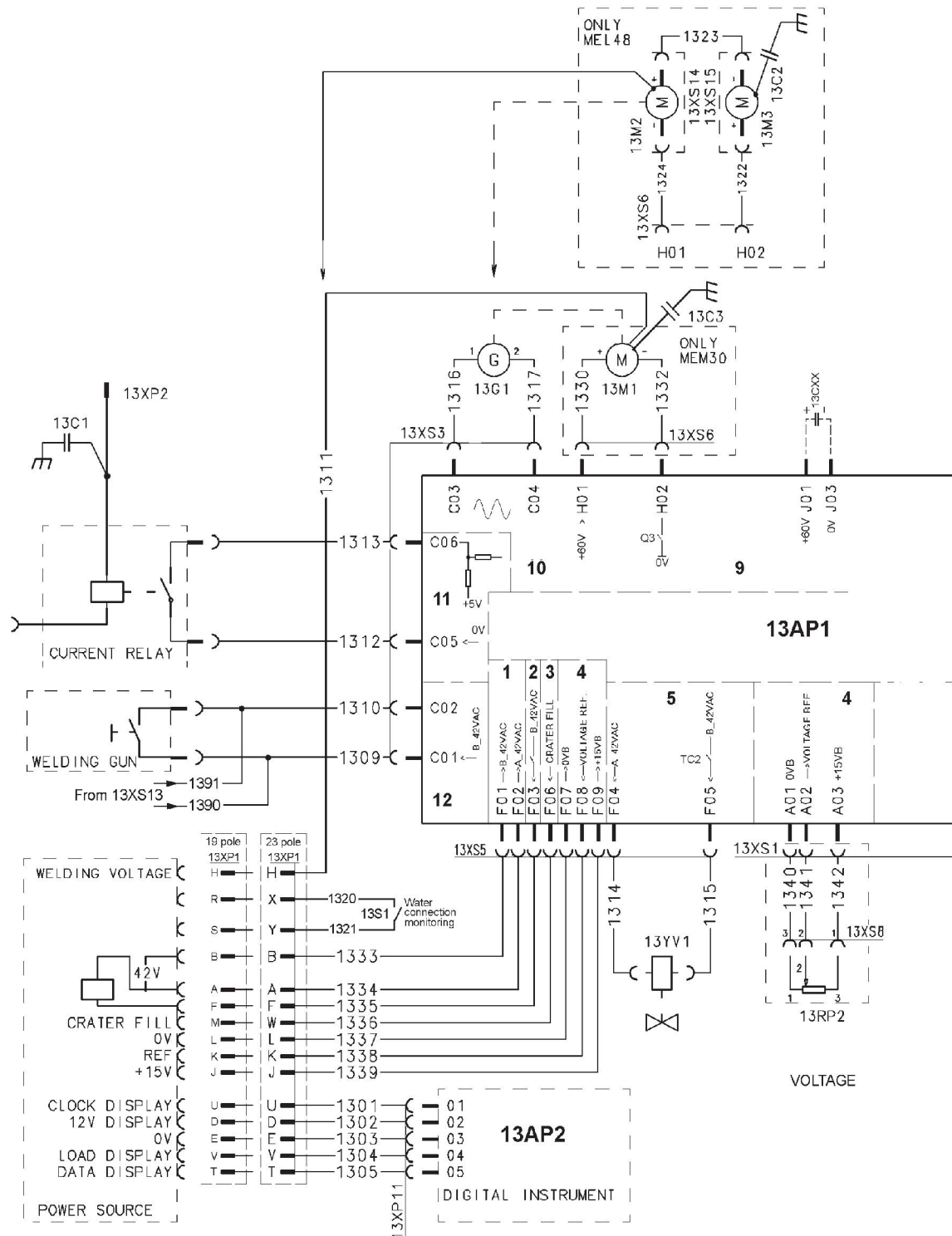


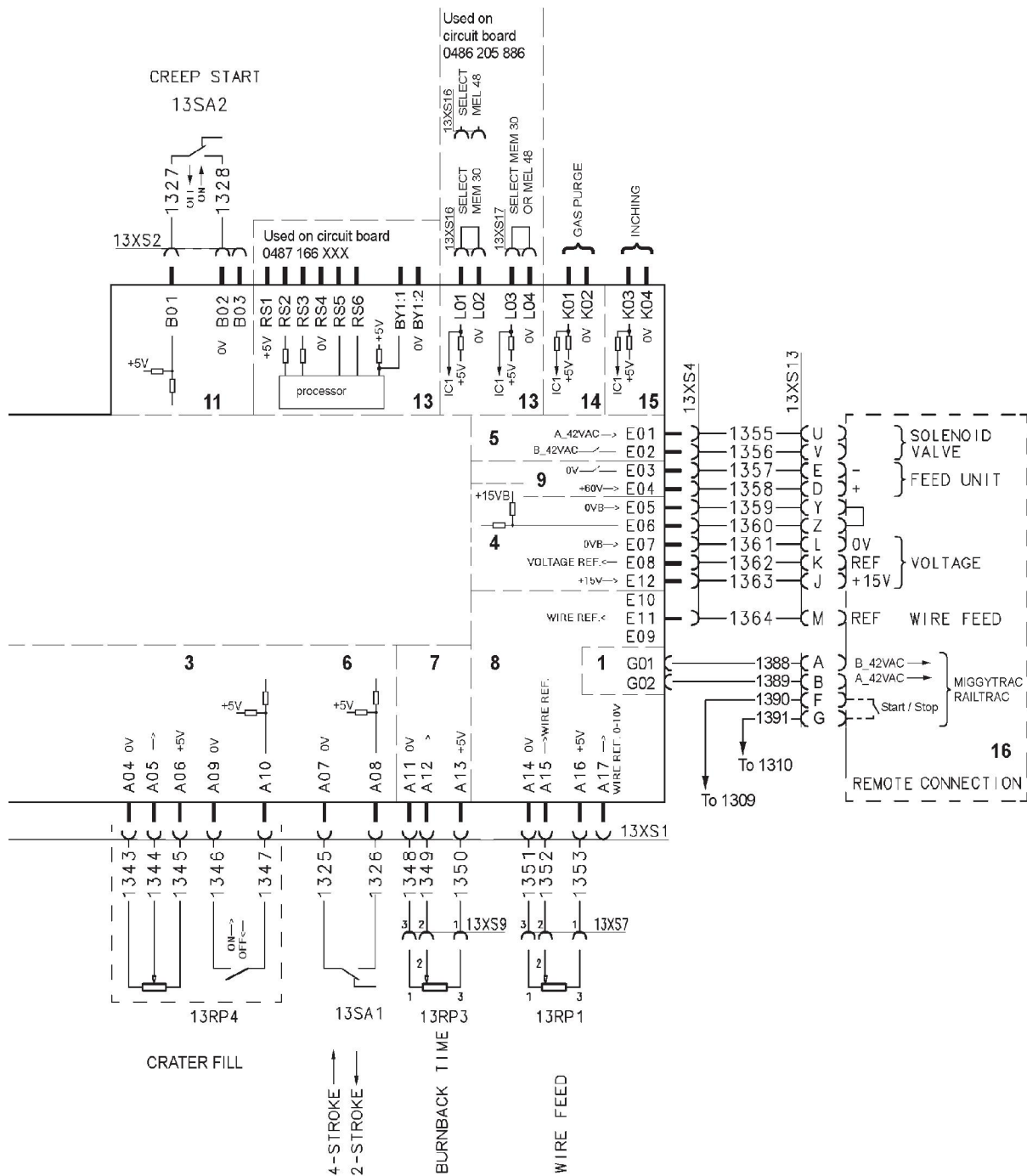
Feed 304 M12 from serial no. 623-xxx-xxxx



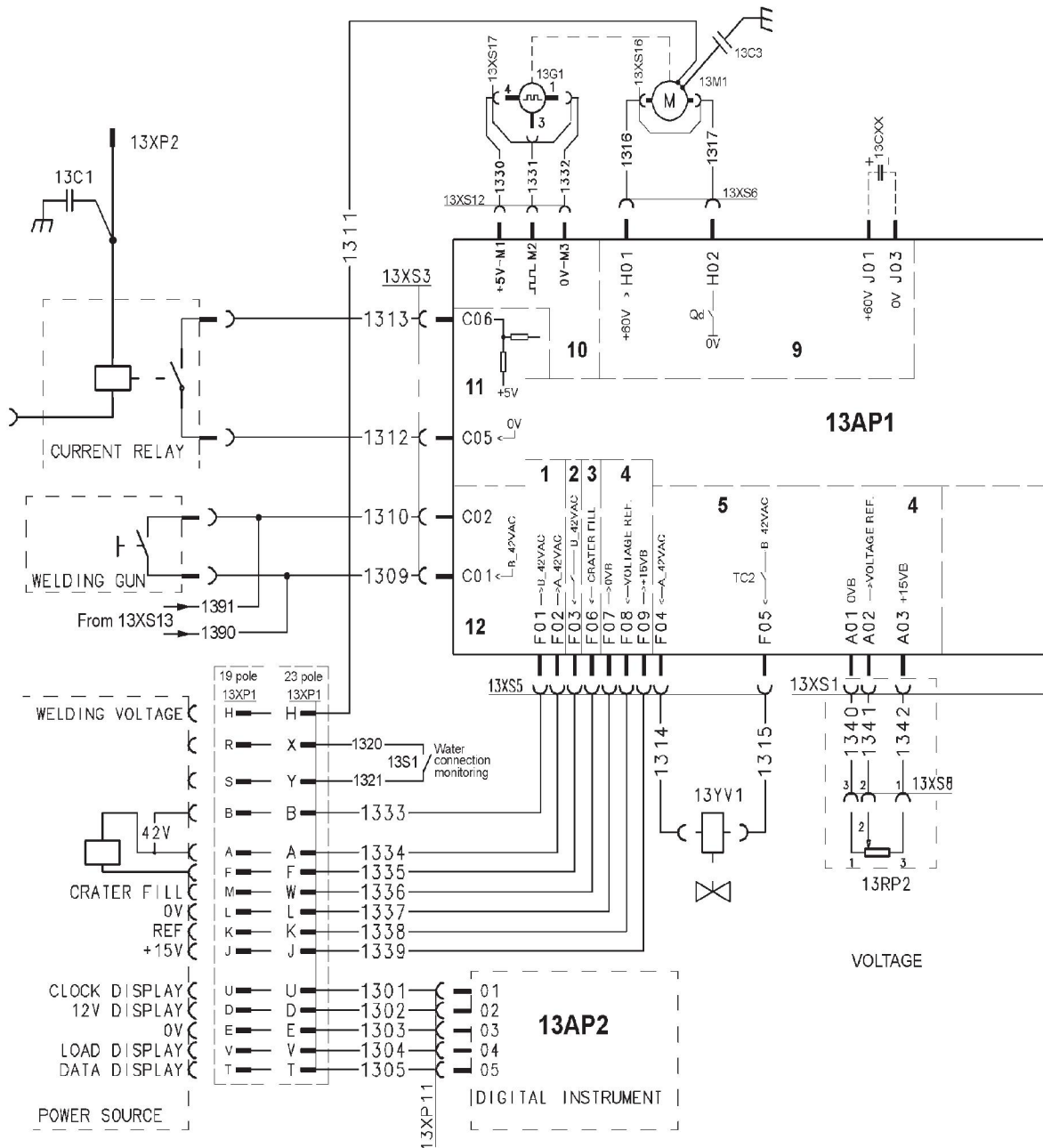


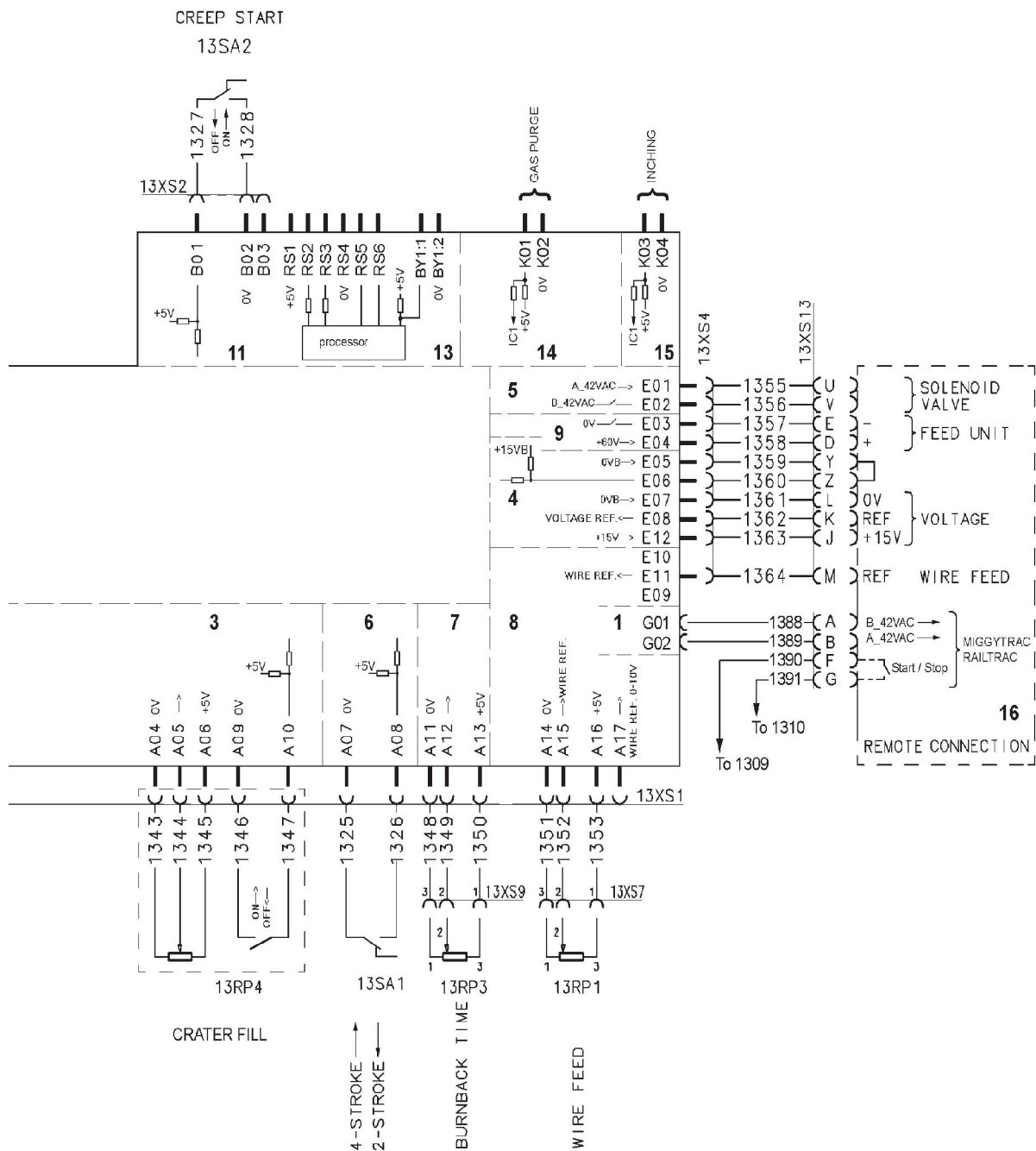
**Feed 304 M13 up to and including serial no. 620-xxx-xxxx,
Feed 484 M13 all serial numbers**



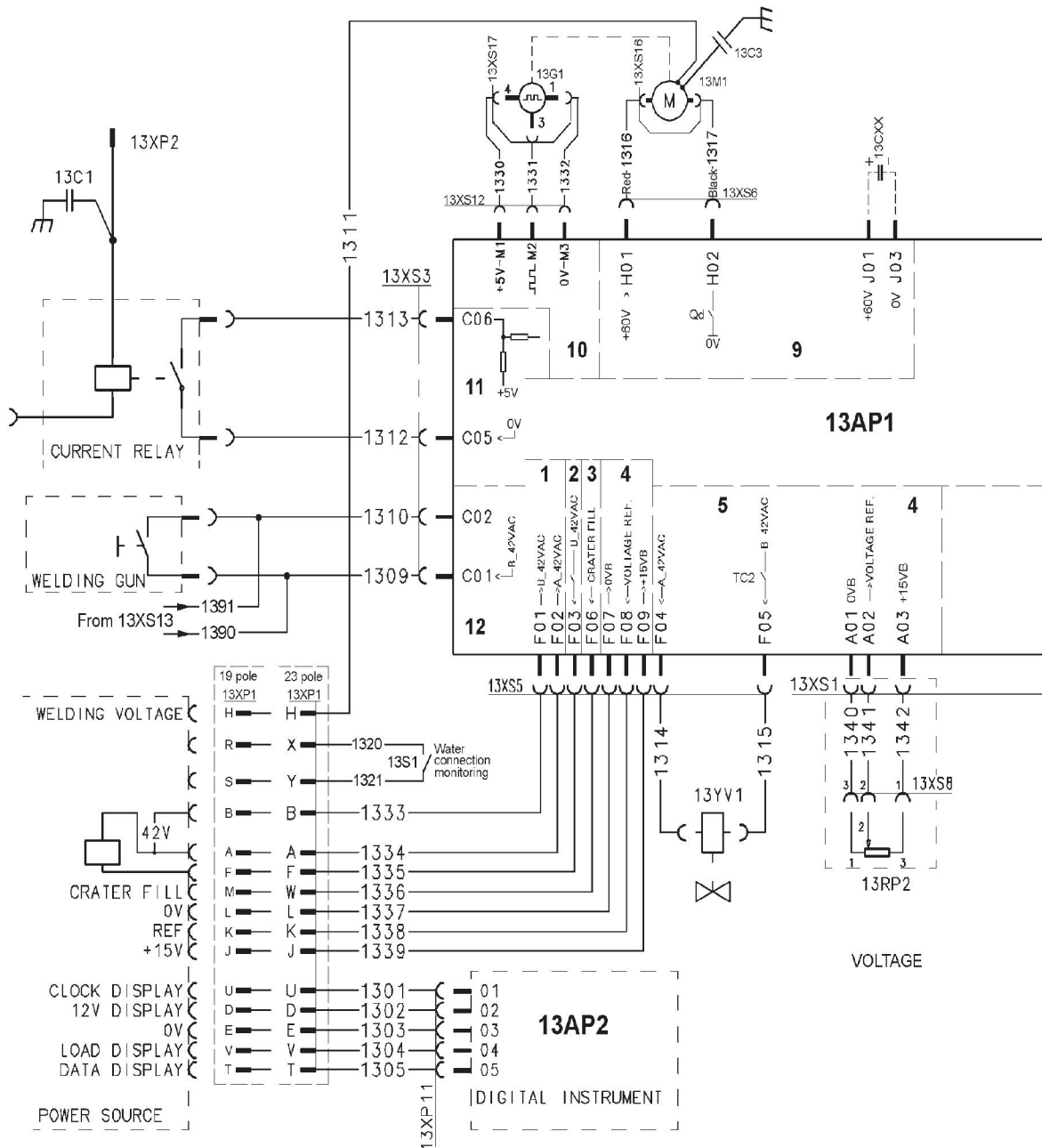


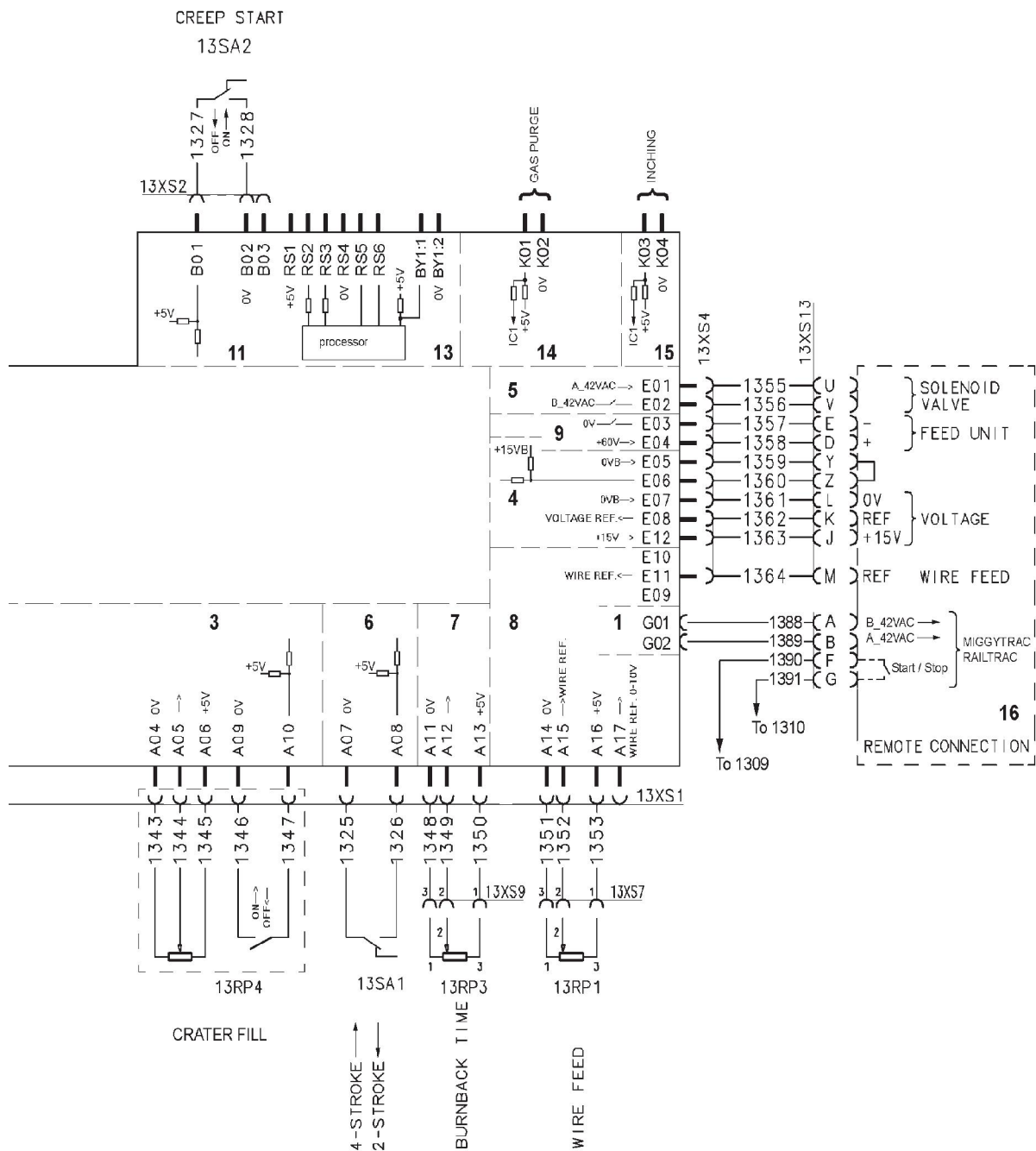
Feed 304 M13 serial no. 917-xxx-xxxx





Feed 304 M13 from serial no. 623-xxx-xxxx





DESCRIPTION OF OPERATION

This description of operation describes the function of circuit boards and other components in the equipment. It is divided into sections, numbered to correspond to the circuit board numbers and divisions into function blocks.

13AP1 Control board

Circuit board identity

The control board has got a machine ID, a hardware ID and a unit type number.

- The machine ID determines which motor type and wire feeder mechanism the board is intended for.
- The hardware ID shows design and type of circuit board.
- The unit type is used for identification on the CAN bus.

For **Feed 304 M12** and **Feed 304 M13** the Machine ID = 46 from serial number 623-xxx-xxxx (due to change of motor).

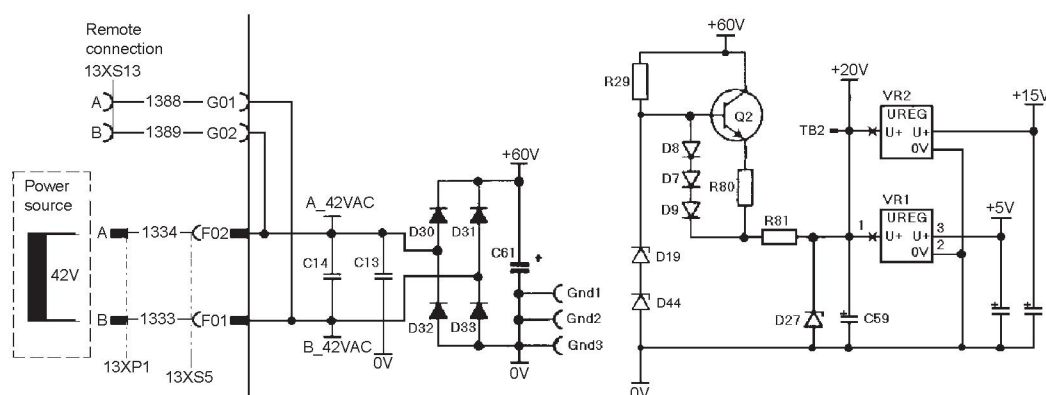
Changes to the circuit board

There are several versions of the control board.

- The first version was used for wire feeders with serial no. 238-xxx-xxxx. This board can be configured for Feed 304 or Feed 484 by jumpers on the board.
- The second and third versions must be configured when the board is produced. See the spare parts list for the correct ordering number.
- From the third version (circuit board print number 487165-003) the supply to the brake circuit is changed, see *"13AP1:9 Motor driving / braking"*, page 26.

Component positions diagram with print number can be found in *"13AP1 Component positions"*, page 32.

13AP1:1 Power supply



The wire feed unit receives 42 V from the control power supply transformer in the welding power source via connector 13XP1.

42 V AC is supplied to the welding torch trigger switch, the gas valve and the contactor. In addition, terminals G01 and G02 supply 42 V to a Miggytrac or a Railtrac unit if this is used together with the wire feed unit.

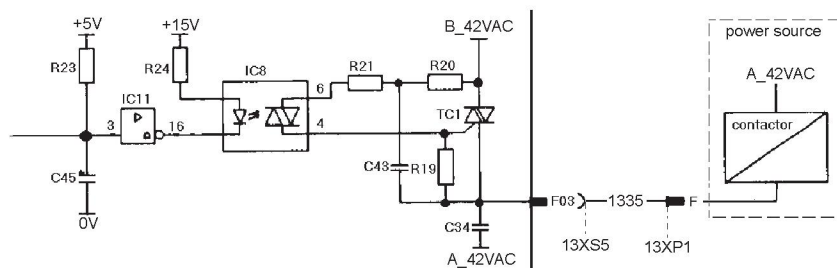
DC power supplies

Diodes D30 – D33 rectify the 42 V supply to 60 V. Capacitor C61 smooths the supply, which is then used to power the wire feeder motor.

Transistor Q2 is a pre-regulator that drops the voltage from 60 V to 20 V. Q2 is current-limited to about 200 mA.

VR1 and VR2 are voltage regulators, producing 5 V and 15 V respectively. The circuit board's microprocessor monitors the voltages. If the 15 V supply falls below 13 V, the wire feed unit is stopped.

13AP1:2 Activation, power source



The start signal is connected to the power source via pin F in connector 13XP1.

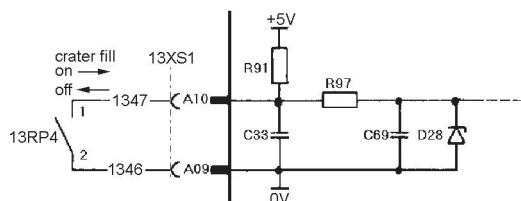
13AP1:3 Crater filling

This function is only used by units with **control panel M13**.

The crater fill function can be used when the feed unit is working with a ESABMig or LAW power source.

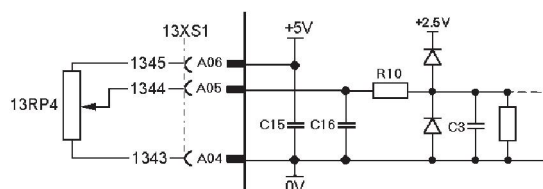
The crater fill function comprises the following modules:

- An input for selecting crater fill function On/Off.



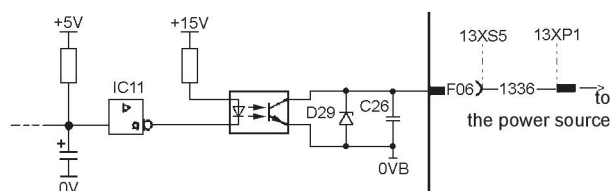
Closing switch 13RP4 activates the function.

- Potentiometer 13RP4 sets the crater fill time.



Times between 0–5.1 s can be set.

- An output, terminal F06, which supplies a signal to the power source when the voltage is to be reduced for crater filling.



The emitter of the transistor in the optocoupler is connected to 0 VB, i.e. to the neutral of the power source voltage reference.

The crater fill function operates as follows:

Phase 1

When in 2-stroke control mode, releasing the welding torch trigger switch reduces the wire feed speed to 84% of its original value. In 4-stroke control mode, crater filling starts when the trigger switch is pressed for the second time. The optocoupler is activated, supplying a signal to the power source that crater filling has started, causing the power source to reduce its output voltage.

Phase 1 continues for a time as determined by the setting of potentiometer 13RP4, up to a maximum of 1.27 s.

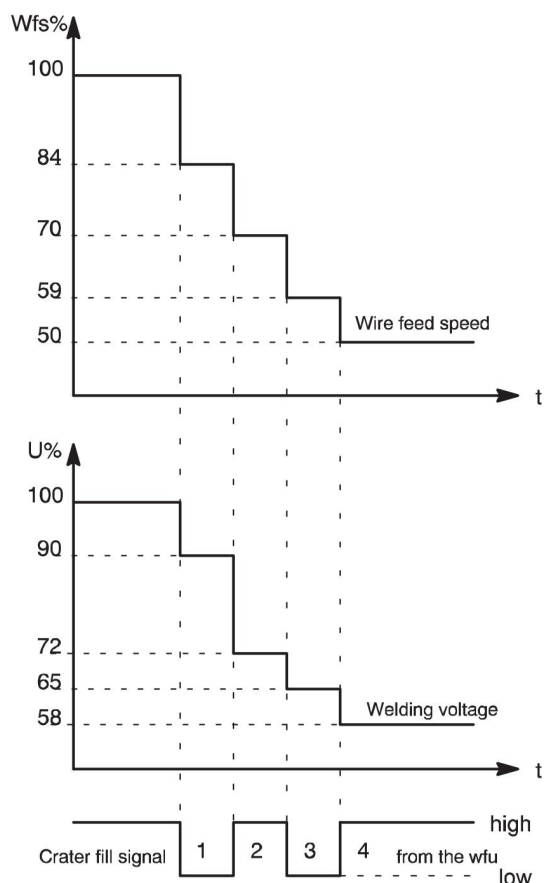
Phase 2

Phase 2 starts when phase 1 finishes. The wire feed unit further reduces the wire speed to 84% of the value during phase 1. The optocoupler turns off, giving the power source a signal further to reduce the voltage. Phase 2 continues also for a maximum of 1.27 s.

Phase 3 and 4

Phases 3 and 4 are repeats of phases 1 and 2. After phase 4, during which wire feed speed is 50% of the original value, welding concludes with the preset burn-back time.

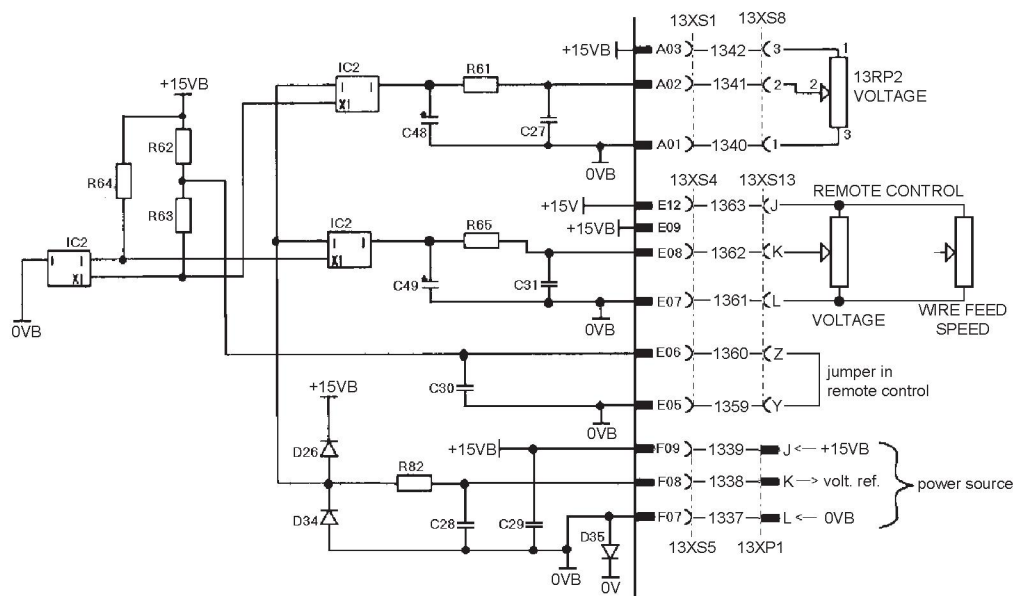
If, when in 4-stroke control mode, the welding torch trigger switch is released while crater filling is in progress, crater filling will be interrupted.



Wire feed speed and welding voltage during crater filling

13AP1:4 Voltage adjustment

This function is only used by units with **control panel M13**.

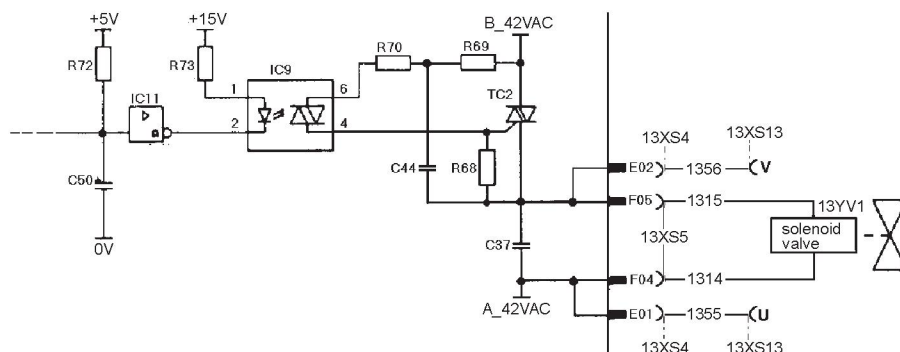


When the wire feed unit is connected to a power source with steplessly variable output voltage, potentiometer 13RP2 on the front of the unit can be used to control the power source output voltage.

The +15 VB and 0 VB voltages are supplied by the power source via connector XP01. The voltage adjustment circuits are not galvanically isolated from the other circuits on circuit board 13AP1.

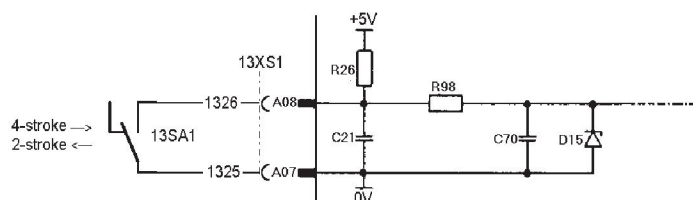
The remote control unit for voltage control is connected to pins J, K, L, Z and Y in connector 13XS13. A link must be fitted between pins Z and Y in order to activate the unit.

13AP1:5 Gas valve



The gas valve is connected to connectors F04 and F05. If an intermediate wire feed unit is being used, its gas valve is connected via pins V and U in the remote control connector (13XS13).

13AP1:6 2-stroke / 4-stroke



2-stroke

When switch 13SA1 is open, 2-stroke control mode is selected. This means that closing the welding torch trigger switch starts the wire feed motor, opens the gas valve and closes the power source contactor.

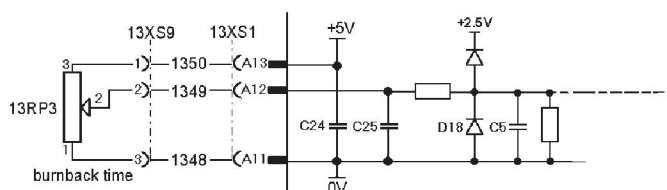
Releasing the switch stops the motor, releases the contactor and closes the gas valve. If crater filling and/or burn-back time are operative, they will be activated before welding ceases.

4-stroke

When switch 13SA1 is closed, 4-stroke control mode is selected. This means that first closure of the trigger switch operates the gas valve, with the wire feed motor starting, and the power source contactor operating, when the trigger switch is released.

Closing the trigger switch for the second time stops the motor and de-energises the contactor in the power source. Releasing the switch closes the gas valve. If crater filling and/or burn-back time are operative, they will be activated before welding ceases.

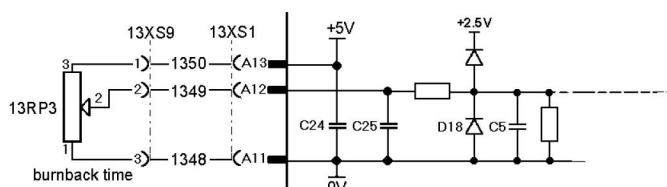
13AP1:7 Burn-back time



The burn-back time is the time from when motor braking starts until the machine contactor opens. It can be adjusted by potentiometer 13RP3 between 0–0.5 s.

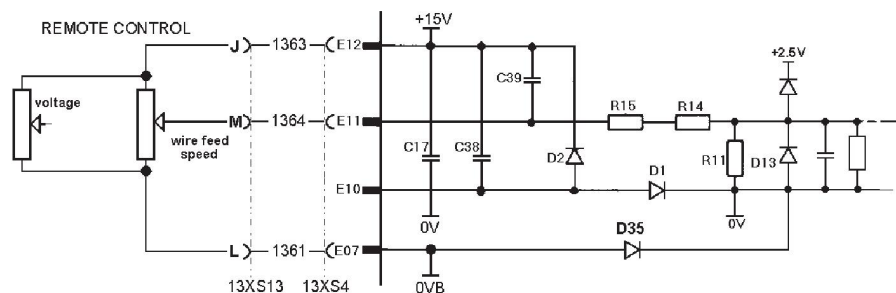
13AP1:8 Wire feed speed

Adjustment on the front panel of the feeder unit



Potentiometer 13RP1 controls the wire feed speed over a range of 1.9–25 m/min.

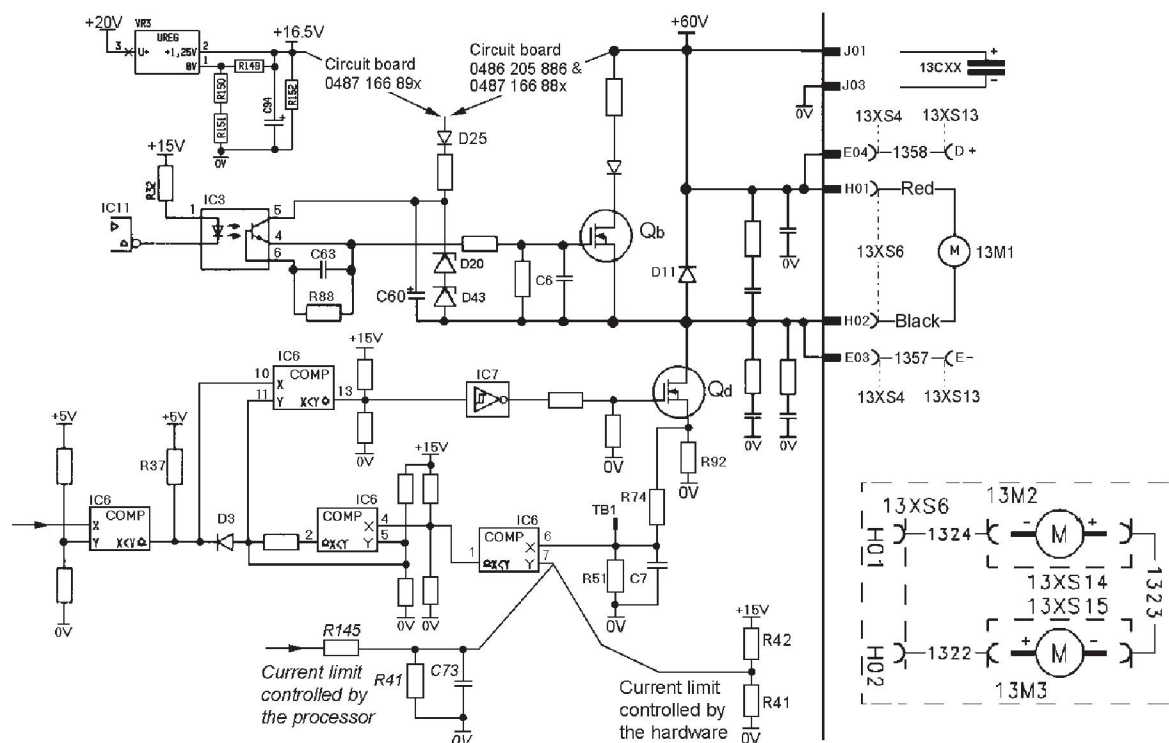
Adjustment using the remote control unit



Wire feed speed reference from the remote control unit

Potentiometer 13RP1 is disengaged when a remote control unit is connected to input 13XS13. The circuit board microprocessor senses that the remote control unit is connected by means of the voltage drop across diode D35. When the remote control unit is connected, the voltage on connector E11 cannot drop below about 0.6 V.

13AP1:9 Motor driving / braking



The motor is powered by the smoothed +60 V supply. Pulse width modulation, controlled by transistor Qd, is used to vary the motor voltage. The pulse frequency is 39 kHz, the old version of the board has a pulse frequency of 12 kHz. The maximum conducting time of the pulses is 99% of the pulse cycle time. Free-wheel diode D11 maintains motor current during the pulse breaks.

Feed 304: At a drive roller speed of 273 r/min, the wire feed speed is 25 m/min.

Feed 484: At a drive roller speed of 166 r/min, the wire feed speed is 25 m/min.

If the wire speed deviates from the set value by more than 1.5 m/min for more than 5 s, wire feed will be stopped. From circuit board, print no. 487165-003 (see "13AP1 Component positions", page 32), the maximum speed deviation is ± 0.9 m/min.

Pins D and E in the remote control unit connector 13XS13 provide power for the motor in an intermediate wire feed unit or for a PKE (push/pull) welding torch. If an intermediate wire feed unit is being used in extreme applications (low mains voltage and long cables), an additional 4700 μ F smoothing capacitor 13CXX must be connected (ordering number 0193 141 048).

Current limit

The voltage drop across resistor R92 provides a signal that is proportional to the motor current. When the current exceeds the current limit, IC6:2 turns off the gate pulse to Qd. When the current drops, Qd conducts again at the next gate pulse. The first version of the control board (0486 205 886) has a fixed current limit at 15.4 A. The current limit is related to the pulse peak current.

The new version of the control board (0487 166 8xx) has a dynamic current limit controlled by the processor.

Feed 304: The current limit is 20 A on starting. When the set speed is achieved, this is reduced to 13 A.

Feed 484: The current limit is 20 A on starting. When the set speed is achieved, this is reduced to 15 A.

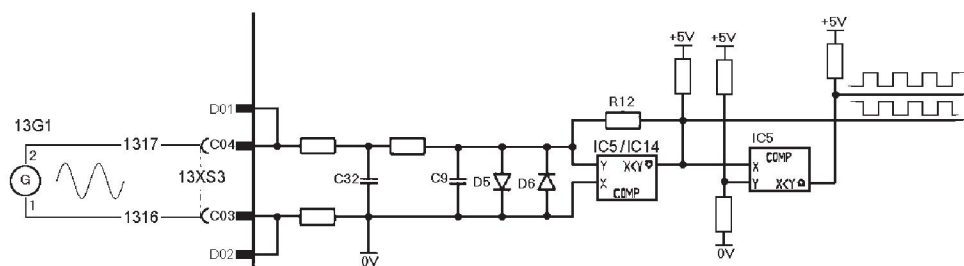
Braking

When the motor starts, capacitor C60 charges up to 15 V, with zener diodes D20 and D43 limiting the voltage across it. Braking is activated by the optocoupler IC3. When the transistor in IC3 is turned on by the LED, 15 V from capacitor C60 is connected to the gate of transistor Qb. Transistor Qb turns on and short circuits the motor through the resistors R85 – R87. The resistors limit the braking current to about 20 A.

13AP1:10 Tachometer input

Feed 304: valid for serial numbers 238-xxx-xxxx, 410-xxx-xxxx, 440-xxx-xxxx, 502-xxx-xxxx and 620-xxx-xxxx.

Feed 484: valid for all serial numbers.



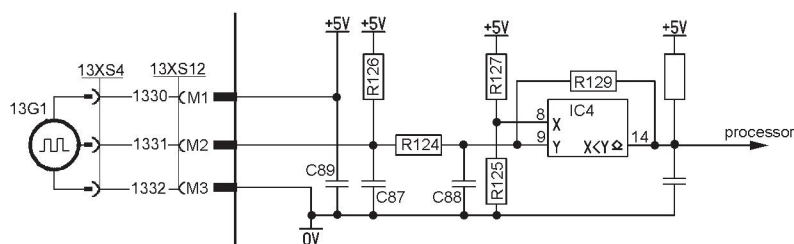
The tachometer 13G1 is fitted inside the motor casing. Comparator IC5 / IC14 converts the sine wave signal from the tachogenerator to a square wave at the same frequency.

Feed 304: At a wire feed speed of 25 m/min, it produces a signal frequency of 667 Hz.

Feed 484: At a wire feed speed of 25 m/min, it produces a signal frequency of 814 Hz.

Pulse generator input

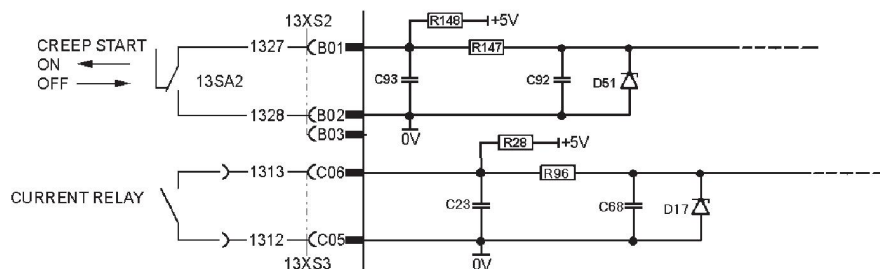
Valid for **Feed 304** from serial number 917-xxx-xxxx.



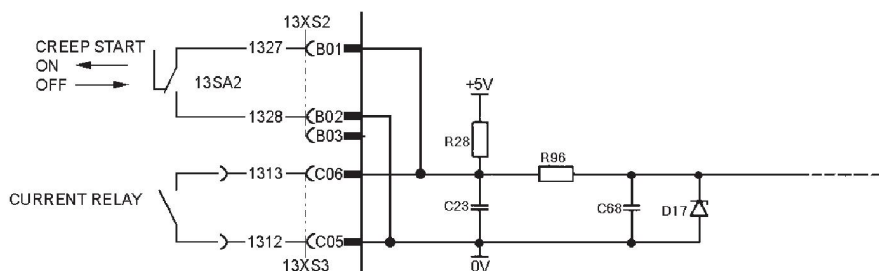
The pulse generator (encoder), 13G1, is fitted to drive motor 13M1. It generates 60 pulses per revolution.

At a wire feed speed of 25 m/min, the pulse generator produces a pulse frequency of 10230 Hz.

13AP1:11 Current relay, Creep start / Normal start



The new version of the control board, 0487 166 8xx



The first version of the control board, 0486 205 886

Current relay

The current relay operates if the welding current exceeds 20 A. If the crater filling function has been selected, it will work only when the current relay has operated.

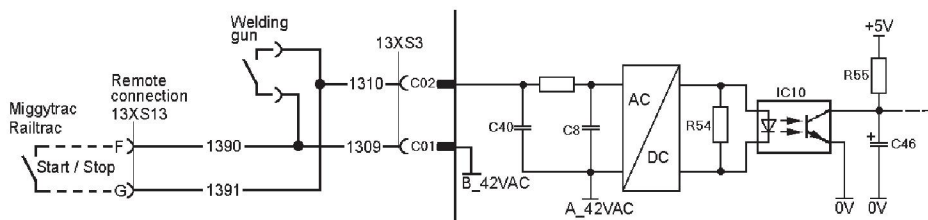
Creep start

The creep start function is activated when switch 13SA2 is open. Creep start means that the motor runs at a speed of 1.9 m/min until the current relay is activated. When the relay operates, the speed increases to the set speed.

The new version of the control board, 0487 166 8xx: If the current relay does not operate within 3 s after starting, the motor speed increases to the set speed in any case.

The first version of the control board, 0486 205 886: If the current relay does not operate within 1 s after starting, the motor speed increases to the set speed in any case.

13AP1:12 Start / Stop



The trigger switch in the welding torch is supplied at 42 V AC. Closing the switch energises optocoupler IC10, pulling down the voltage across C46.

When the Miggytrac or the Railtrac is used, the start signal to the feed unit is supplied by the remote input.

The new version of the control board, 0487 166 8xx: When wire feed starts, the power source generates welding voltage. If this is not followed by a flow of welding current within 3 s, the power source shuts down the welding power supply, although wire feed continues until the welding torch switch is released.

The first version of the control board, 0486 205 886: When wire feed starts, the power source generates welding voltage.

13AP1:13 Processor, Starting sequence, Motor type

Processor

The processor stores the machine program. It monitors the power supply voltages. If the voltages drop to too low a level, wire feed is stopped, as described in section 13AP1:1 above.

The processor also monitors speed. If the wire speed deviates from the set value by more than 1.5 m/min for more than 5 s, wire feed will be stopped.

Starting sequence, circuit board 0487 166 8xx

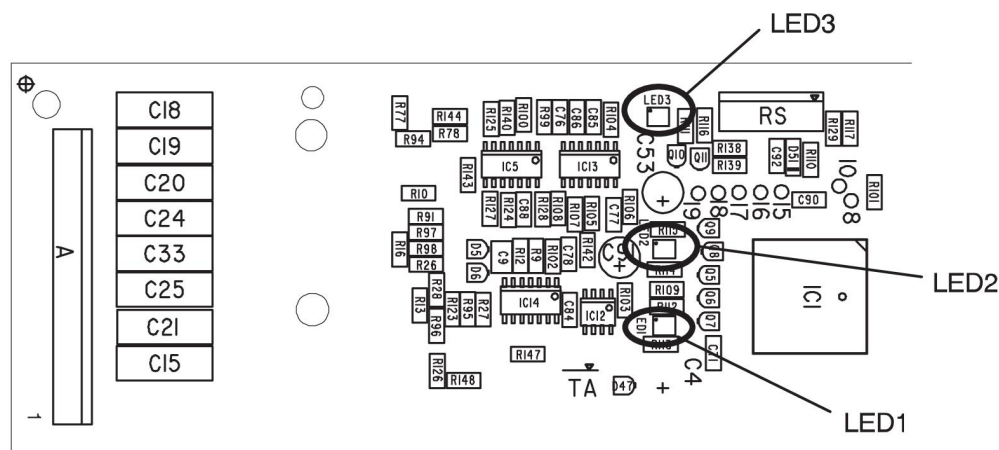
The circuit board displays the starting sequence from power-up.

1. LED1 lights red during 5 s. The software is in the boot mode.
2. LED1, LED2 and LED3 lights green. The software has started the application mode.
3. LED1 flashes continuously with a green light. The board is initiated, and the wire feed unit is in the application program.

The starting sequence above applies to boards with software version 2.00A and later.

Boards with software version 0.27A have LED indication as described in paragraphs 2 and 3 above.

Boards with software version prior to 0.27A have no LED indication. The software version is indicated by a label on the processor, IC1.



LEDs on circuit board 13AP1

Motor type

The wire feeder mechanisms MEM 30 (for Feed 304) and MEL 48 (for Feed 484) need different software for the motors. On the first version of the control board (0486 205 886) the motor type is selected by the hardware. On the new versions the motor type is selected by the software.

The new version of the control board, 0487 166 8xx:

When the board is tested, the software is set up for one of the motor types.

Feed 304 before serial no. 917-xxx-xxxx, motor with tachometer.

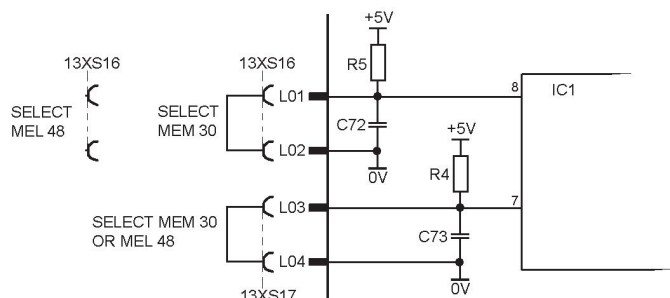
Feed 304 from serial no. 917-xxx-xxxx, motor with pulse generator.

Feed 484 all serial numbers, motor with tachometer.

See the spare parts list for ordering numbers.

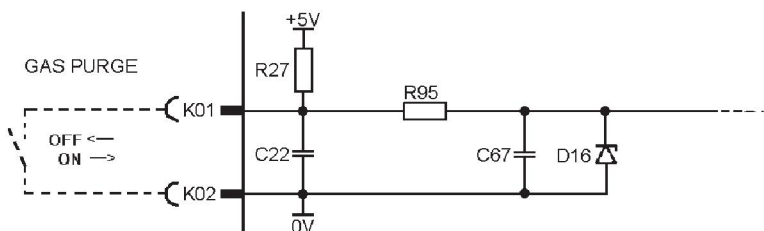
The first version of the control board, 0486 205 886:

To enable the processor IC1, to select the correct software for the ESABFeed 30-4, connection L01 must be linked to L02 and L03 to L04. For the ESABFeed 48-4 there is a link only between L03 and L04.



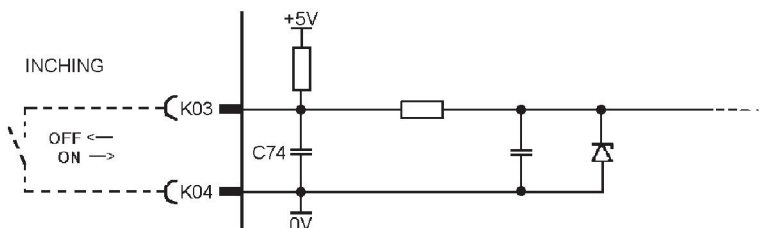
Selection of motor type, circuit board 0486 205 886

13AP1:14 Gas purge



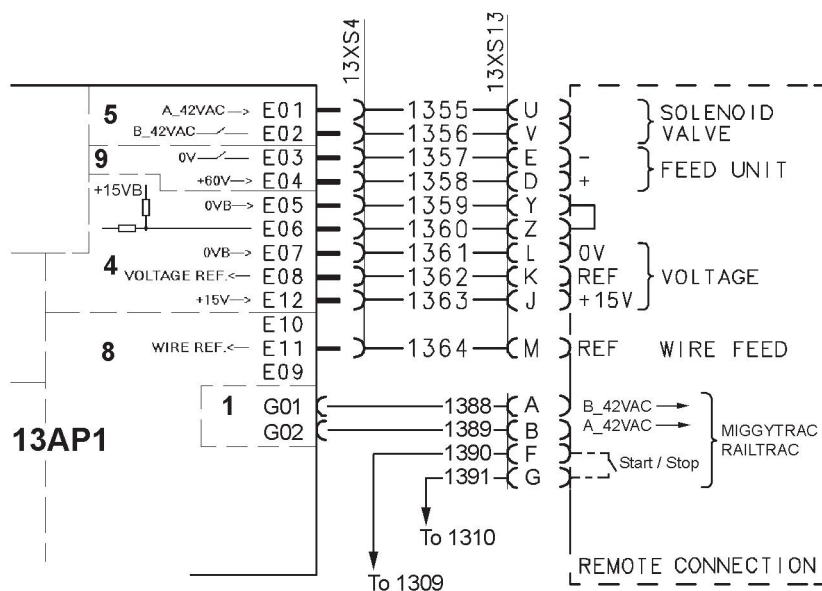
The circuit board has an input for gas purge, this is not used.

13AP1:15 Inching



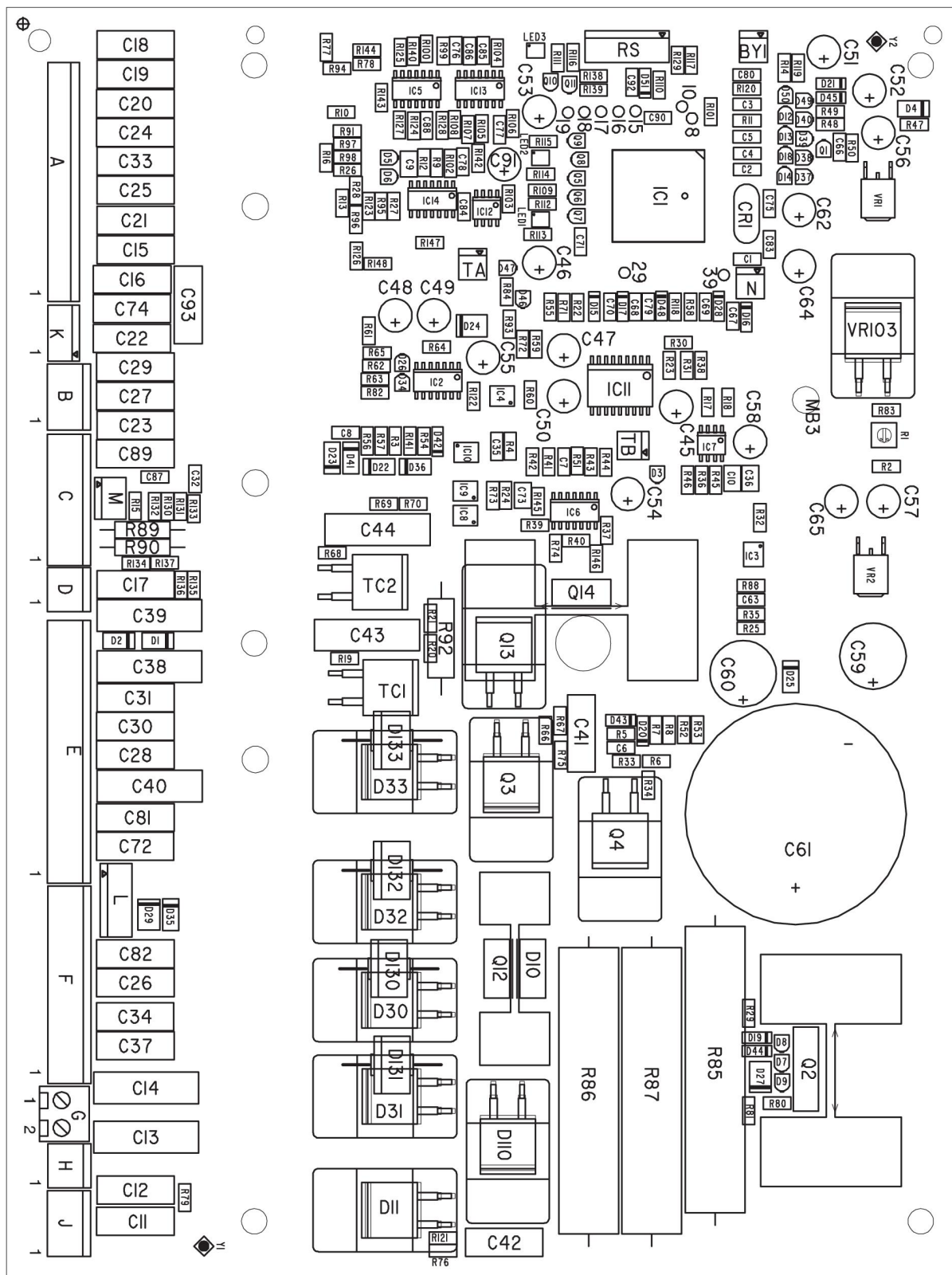
The circuit board has an input for inching, this is not used.

13AP1:16 Remote control input

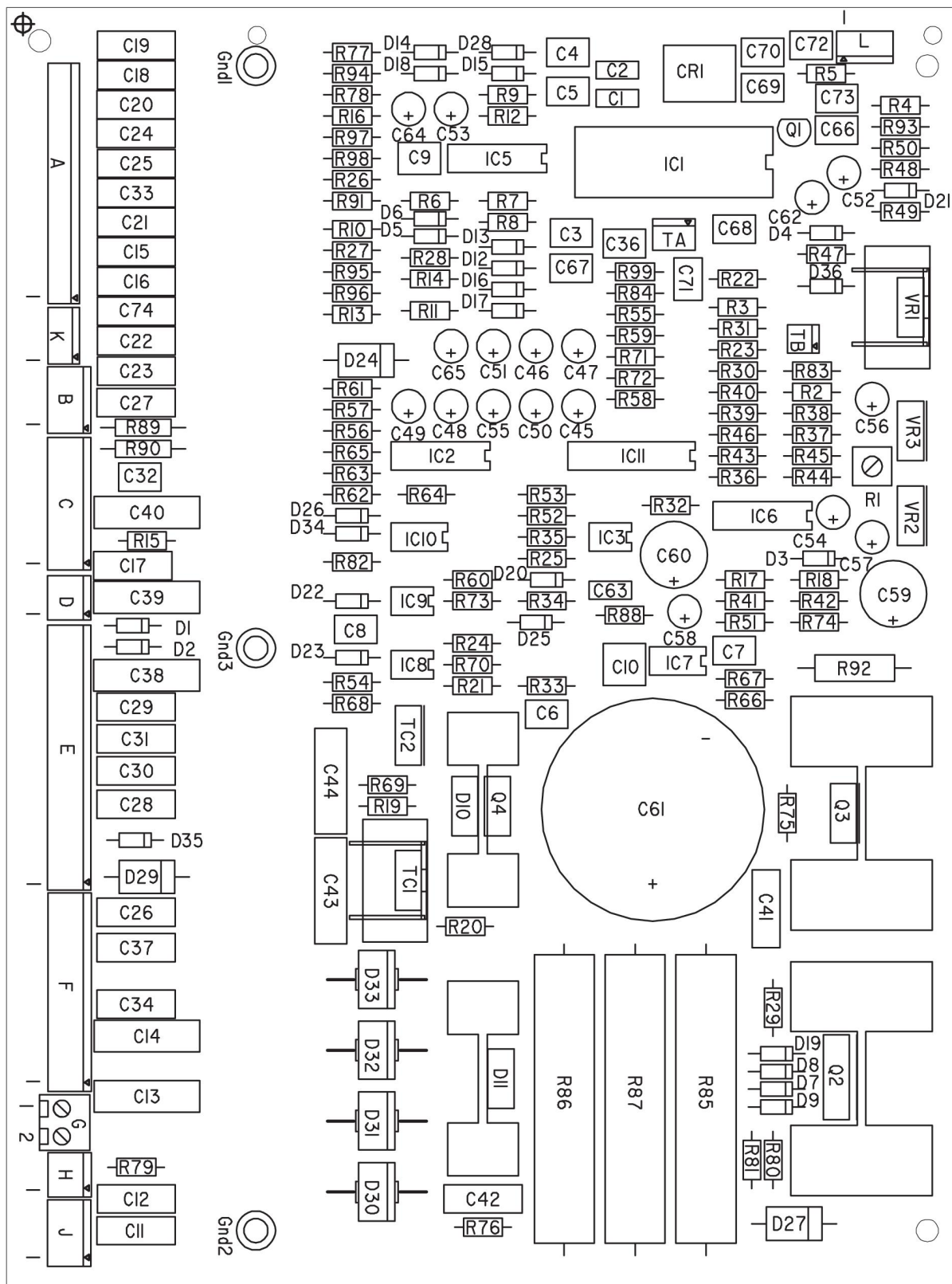


The following items can be connected to the remote control input:

- Wire feed speed remote control unit: pins M, J and L in 13XS13.
- Welding voltage remote control unit, if the power source is of thyristor-controlled or transistor-controlled type: pins J, K, L, Z and Y in 13XS13.
- MEK 25 intermediate wire feeder unit.
- Equipment for mechanized welding: Miggytrac or Railtrac.



The second version of the control board, 0487 166 88x



The first version of the control board, 0486 205 886

13AP2 Digital display

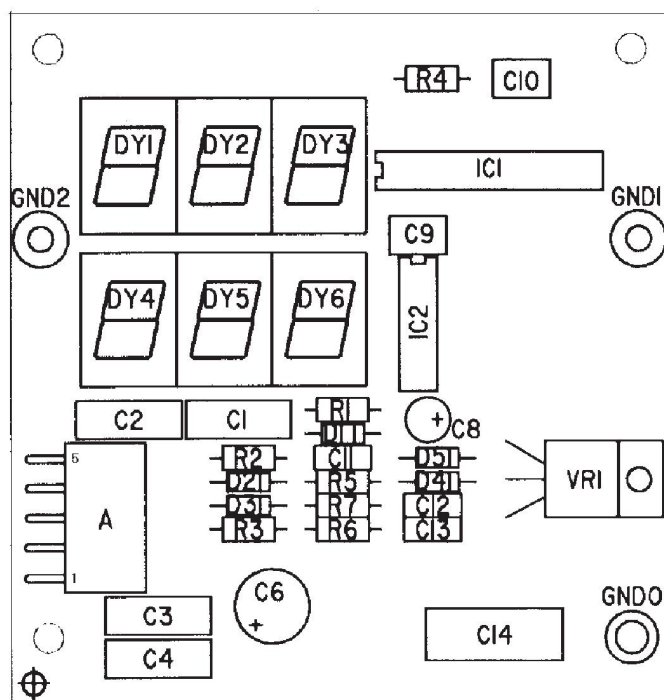
The digital display unit is used for wire feed units with **control panel M13**.

The instrument can be connected either to the wire feeder unit or to the power source. Connection to the power source requires a connection kit. The instrument can be used only when the wire feeder unit is connected to one of the following welding power sources:

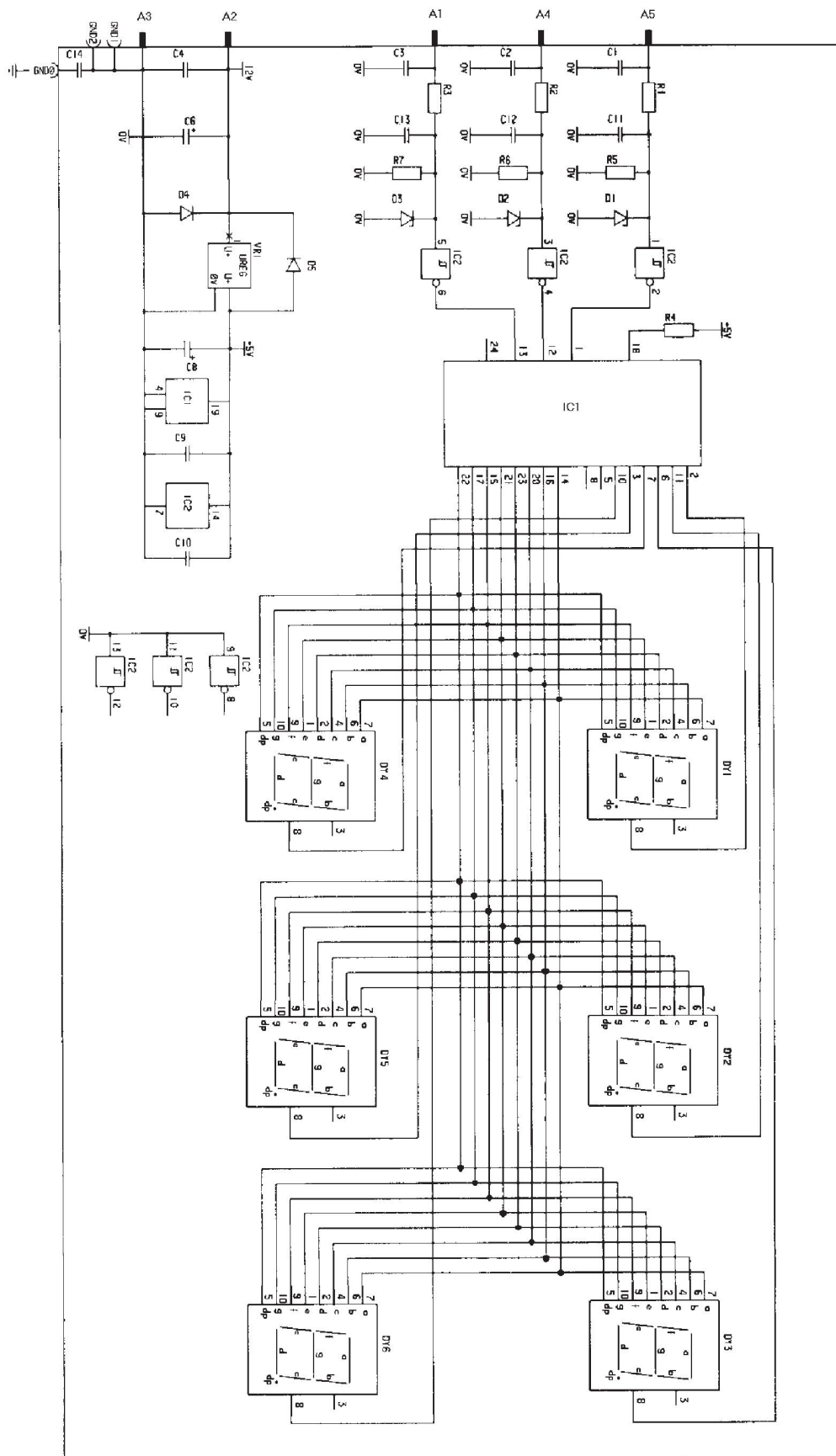
- Mig 402c, Mig 502c, Mig 652c
- Mig 400t, Mig 500t
- ESABMig 400t, ESABMig 500t
- LAW 420, LAW 520
- LAW 410, LAW 510
- LAW 400, LAW 500

The display board has two rows of 7-segment displays, with three displays in each row. The upper row displays arc voltage and the lower displays the welding current.

Data is transmitted serially to the display board from the power source, via inputs A1, A4 and A5. A smoothed 12 V DC power supply is supplied to the board via inputs A2 and A3 from the welding power source. Voltage regulator VR1 produces a 5 V output.



Component positions, display board 13AP2



Circuit diagram, display board 13AP2

SERVICE INSTRUCTIONS



CAUTION!

STATIC ELECTRICITY can damage circuit boards and electronic components.

- Observe precautions for handling electrostatic sensitive devices.
- Use proper static-proof bags and boxes.



ESD

What is ESD?

A sudden transfer or discharge of static electricity from one object to another. ESD stands for Electrostatic Discharge.

How does ESD damage occur?

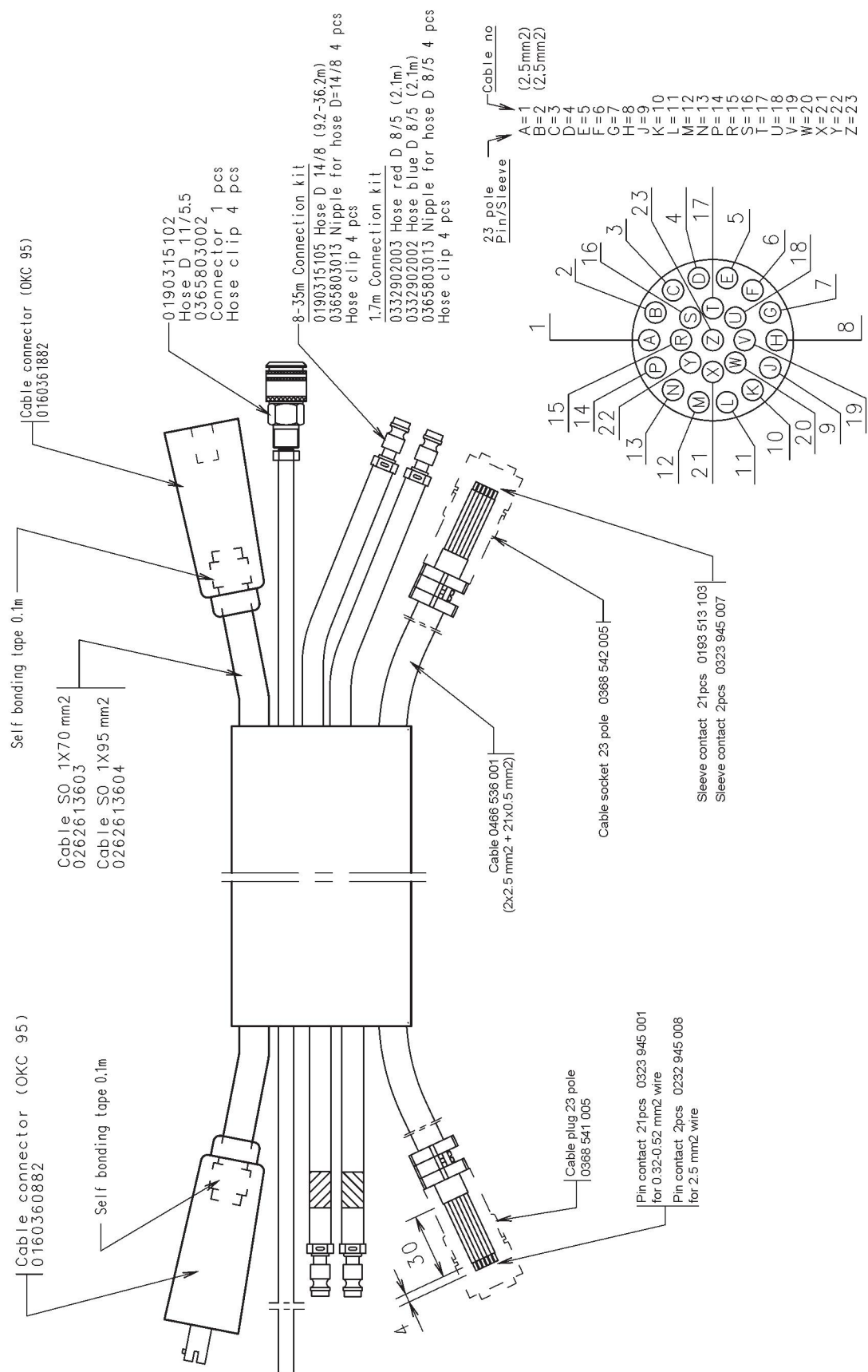
ESD can cause damage to sensitive electrical components but is not dangerous to people. ESD damage occurs when an ungrounded person or object with a static charge comes into contact with a component or assembly that is grounded. A rapid discharge can occur, causing damage. This damage can take the form of immediate failure, but it is more likely that system performance will be affected and the component will fail prematurely.

How do we prevent ESD damage?

ESD damage can be prevented by awareness. If static electricity is prevented from building up on you or on anything at your work station, then there cannot be any static discharges. Nonconductive materials (e.g. fabrics), or insulators (e.g. plastics) generate and hold static charge, so you should not bring unnecessary nonconductive items into the work area. It is obviously difficult to avoid all such items, so various means are used to drain off any static discharge from persons to prevent the risk of ESD damage. This is done by simple devices: wrist straps, connected to ground, and conductive shoes.

Work surfaces, carts and containers must be conductive and grounded. Use only antistatic packaging materials. Overall, handling of ESD-sensitive devices should be minimized to prevent damage.

Spare parts for connection sets, 23 pole



SPARE PARTS

The spare parts list is published in a separate document that can be downloaded from the Internet: www.esab.com

File name	Product
0459 161 990	Feed 304 M12 / M13 and Feed 484 M12 / 13



A WORLD OF PRODUCTS AND SOLUTIONS.



For contact information visit esab.com

ESAB AB, Lindholmsallén 9, Box 8004, 402 77 Gothenburg, Sweden, Phone +46 (0) 31 50 90 00

<http://manuals.esab.com>

